

INTRODUCTION TO DATA SCIENCE AND PROGRAMMING

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THANKS TO LIAM POWER

A REAL-LIFE EXAMPLE

- What if you want to execute this code block twice?

```
name = input("What is your name?")
```

A REAL-LIFE EXAMPLE

- What if you want to execute this code block twice?
- What about three times?

```
name = input("What is your name?")  
name = input("What is your name?")
```

A REAL-LIFE EXAMPLE

- What if you want to execute this code block twice?
- What about three times?
- What about four times? Or more?

```
name = input("What is your name?")  
name = input("What is your name?")  
name = input("What is your name?")
```

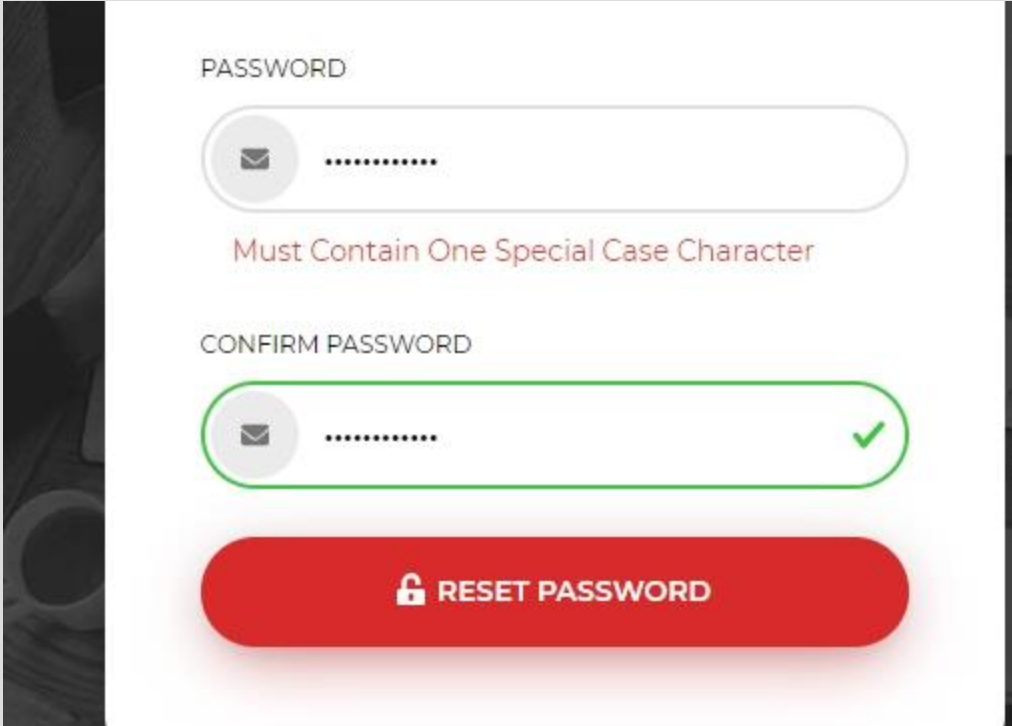
ITERATION

WHAT IS ITERATION?

- **Iteration** means **repeating** a piece of code **multiple times**
- 2 main types:
 1. Repeating until a **condition is met** – *While* loop
 2. Repeating for a **specific number of times** – *for* loop

WHILE LOOPS

- A **while loop** is used when you want to keep repeating until a **certain condition** becomes **false (boolean)**
- For example, you want to repeatedly ask to set a password under a certain length



The image shows a password reset form. It has two input fields: 'PASSWORD' and 'CONFIRM PASSWORD'. The 'PASSWORD' field has a red error message below it: 'Must Contain One Special Case Character'. The 'CONFIRM PASSWORD' field has a green checkmark icon on the right. Below the fields is a red button with a lock icon and the text 'RESET PASSWORD'.

PASSWORD

Must Contain One Special Case Character

CONFIRM PASSWORD

RESET PASSWORD

```
password = input("Create a password (at least 6 characters)")  
  
while len(password) < 6:  
    print("password is too short! Try again")  
    password = input("Create a password (at least 6 characters)")  
  
print("Successfully created password!")
```

WHILE LOOPS

FOR LOOPS

- What if you wanted to print the numbers from 1 to 5?
- `Range(start, stop)` - up to but **not including stop!**
- `Range(1, 5) = [1, 2, 3, 4]`
- `Range(10, 15) = [10, 11, 12, 13, 14]`
- `i` is a **variable** that changes value after each iteration

```
for i in range(1, 6): # [1, 2, 3, 4, 5]
    print(str(i))

# 1
# 2
# 3
# 4
# 5
```

FOR LOOPS

- What if you wanted print your name 3 times?

```
myName = "Brendan"
for i in range(0, 3): # [0, 1, 2]
    print(myName)

# Brendan
# Brendan
# Brendan
```

BREAK

- What if you want to end a loop early?
 - Generate numbers from 100 to 200, and **stop** at a **prime number**

```
for i in range(100, 200): # [100, 101, ..., 199]
    print(str(i))
    if isPrimeNumber(i):
        break
```

CONTINUE

- What if you want to skip an iteration but keep the loop running?
 - Print all numbers from 100 to 200 **except prime numbers**

```
for i in range(100, 200): # [100, 101, ..., 199]
    if isPrimeNumber(i):
        continue
    print(str(i))
```

PRACTICE

Part 2